Modified Denavit–Hartenberg Modeling and Numerical Inverse Kinematics of the Open Duck Mini Robot

ROSE5710: Fundamental Robotics and Systems
Assignment 1: Kinematics, First Term 2026–2027

***Abstract*—This report develops modified Denavit–Hartenberg models for the five-joint left and right legs and the four-joint head–neck chain of the Open Duck Mini robot. Joint directions and fixed site transforms are derived from the supplied geometry and checked against the distributed MuJoCo model. An inverse-kinematics solver combines a weighted position–orientation objective with central finite differences, a fixed gradient-descent step, and projection onto joint limits. Forward kinematics is evaluated at 300 seeded random configurations. Inverse kinematics is evaluated at 1,515 chain poses over five complete trajectory cycles. Every recorded position and orientation error satisfies the assignment tolerances. The measurements establish numerical agreement with the supplied kinematic model; they do not constitute physical calibration or dynamic tracking experiments.**

***Index Terms*—Bipedal robots, forward kinematics, modified Denavit–Hartenberg parameters, numerical inverse kinematics, MuJoCo.**

I. Introduction

The assignment requires analytical forward kinematics (FK) and numerical inverse kinematics (IK) for two legs and a head–neck chain [1]. The model must reproduce both site position and orientation, including the final fixed transform from the last joint to the foot or head site. The distributed Open Duck Mini XML model supplies the independent reference geometry [2]; MuJoCo computes its reference body and site poses [3].

The report gives the coordinate convention, all three modified Denavit–Hartenberg (MDH) tables and joint signs, the required optimization algorithm, and the measured FK/IK comparisons. Particular care is taken with mirrored joint directions, offsets along parallel axes, and the distinction between a solver stopping condition and independent acceptance against the reference model.

II. Modified Denavit–Hartenberg Model

A. Coordinate and transform conventions

Column vectors are used. The homogeneous transform ${}^{i-1}\mathbf{T}_i$ maps coordinates from frame i to frame $i-1$. For row i, the prescribed MDH order is

$$\mathbf{A}_i = \mathbf{R}_x(\alpha_{i-1})\mathbf{T}_x(a_{i-1})\mathbf{R}_z(\theta_i)\mathbf{T}_z(d_i). \tag{1}$$

Unlike the standard DH indexing, the revolute axis of joint i is z_i in this convention; x_{i-1} follows a common normal to z_{i-1} and z_i. Intersecting axes permit a zero common-normal length. Writing $c_\theta = \cos\theta_i$, $s_\theta = \sin\theta_i$, $c_\alpha = \cos\alpha_{i-1}$, and $s_\alpha = \sin\alpha_{i-1}$ gives

$$\mathbf{A}_i = \begin{bmatrix} c_\theta & -s_\theta & 0 & a_{i-1} \\ c_\alpha s_\theta & c_\alpha c_\theta & -s_\alpha & -d_i s_\alpha \\ s_\alpha s_\theta & s_\alpha c_\theta & c_\alpha & d_i c_\alpha \\ 0 & 0 & 0 & 1 \end{bmatrix}. \tag{2}$$

For a chain with n actuated joints and one fixed site row,

$$\theta_i = s_i q_i + \theta_i^0, \quad i = 1, \ldots, n, \tag{3}$$

$${}^{B}\mathbf{T}_S(\mathbf{q}) = {}^{B}\mathbf{T}_0 \prod_{i=1}^{n+1} \mathbf{A}_i, \tag{4}$$

where row $n+1$ contains no joint variable. The factors are accumulated from left to right in increasing row order. All distances in the tables are in metres and angles are in radians. The code stores rows as $[\alpha, a, \theta^0, d]$; the printed tables use the assignment order $[\alpha, a, d, \theta]$.

B. Left and right leg chains

Define the base-to-yaw planar distance and angle by

$$r = \sqrt{0.019^2 + 0.035^2} = 0.03982461550348 \text{ m}, \tag{5}$$

$$\beta = \operatorname{atan2}(19, 35) = 0.49734271634682. \tag{6}$$

The supplied base-to-frame-zero transforms are retained:

$${}^{B}\mathbf{T}_{0,L} = \mathbf{R}_z(\pi/2 + \beta), \tag{7}$$

$${}^{B}\mathbf{T}_{0,R} = \mathbf{R}_z(-\pi/2 - \beta). \tag{8}$$

Here rotations are understood as homogeneous transforms with zero translation. The signs relating MuJoCo joint coordinates to MDH angles are

$$\mathbf{s}_L = [1, -1, -1, -1, -1]^\mathsf{T}, \tag{9}$$

$$\mathbf{s}_R = [1, -1, 1, -1, -1]^\mathsf{T}. \tag{10}$$

Tables I and II give all joint and site rows. At zero joint angles, both legs use the MDH directions $z_1 = +Z$, $z_2 = +X$, and $z_3 = z_4 = z_5 = +Y$. The first two MuJoCo joint directions are $+Z$ and $-X$. The three pitch directions are $(-Y, -Y, -Y)$ on the left and $(+Y, -Y, -Y)$ on the right. These directions explain the different signs of the third joint.

The first two MDH origins are chosen at the yaw/roll and roll/pitch axis intersections rather than at arbitrary CAD body

TABLE I
MODIFIED D–H PARAMETERS FOR THE LEFT LEG. LENGTHS a, d ARE IN METRES; ANGLES α, θ, q ARE IN RADIANS. THE STARRED FINAL ROW IS THE FIXED SITE TRANSFORM.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 - \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	0.07415	$-q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	-0.01975	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

TABLE II
MODIFIED D–H PARAMETERS FOR THE RIGHT LEG. LENGTHS a, d ARE IN METRES; ANGLES α, θ, q ARE IN RADIANS. THE STARRED FINAL ROW IS THE FIXED SITE TRANSFORM.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	r	-0.0000741	$q_1 + \pi + \beta$
2	$\pi/2$	0	-0.01606	$-q_2 + \pi/2$
3	$\pi/2$	0	-0.07415	$q_3 + \pi$
4	0	0.07865	0	$-q_4$
5	0	0.07865	0.01735	$-q_5 - \pi/2$
6*	$\pi/2$	0.0005	-0.036225	0

origins. The effective roll-axis height follows directly from the XML offsets:

$$45.9409 - 46.015 = -0.0741 \text{ mm}. \tag{11}$$

Translations along the three parallel pitch axes commute with their joint rotations and may be combined at a selected origin. This yields the supplied final lateral distances of 19.75 mm on the left and 17.35 mm on the right. A completely mirrored table would therefore be incorrect. The final fixed row restores the foot-site orientation, which is the identity in the zero pose.

C. Head and neck chain

The supplied neck anchor is already represented in the initial transform:

$$^{B}\mathbf{T}_{0,H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{p}_c \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}, \qquad \mathbf{p}_c = \begin{bmatrix} 0.001 \\ 0.01915 \\ 0.0900009 \end{bmatrix} \text{m}. \tag{12}$$

The four zero-pose joint axes are $[-Y, -Y, +Z, -X]$, matching the MuJoCo positive directions, so $\mathbf{s}_H = [1, 1, 1, 1]^\mathsf{T}$. Table III contains the corresponding five rows.

The base-to-neck distance labelled 92.021 mm in the handout is already contained in $\mathbf{p}_c$ and is not added a second time. The neck-to-head-pitch separation is 66 mm. The effective vertical separation from the head-pitch axis to the roll axis is 37.85 mm, and the net roll-axis offset is 5 mm.

D. Analytical zero-pose check

Let $d_{1x}, d_{1y}, d_4, \ldots, d_{9z}$ denote the leg dimensions in Fig. 2 of the handout, converted to metres. The MDH construction

TABLE III
MODIFIED D–H PARAMETERS FOR THE HEAD. LENGTHS a, d ARE IN METRES; ANGLES α, θ, q ARE IN RADIANS. THE STARRED FINAL ROW IS THE FIXED SITE TRANSFORM.

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	$\pi/2$	0	0	$q_1 + \pi/2$
2	0	0.066	0.01905	$q_2 - \pi/2$
3	$-\pi/2$	0	0.03785	$q_3 + \pi/2$
4	$-\pi/2$	0	-0.005	q_4
5*	$\pi/2$	0	0.04245	$-\pi/2$

TABLE IV
ANALYTICAL AND XML ZERO-POSE POSITIONS IN THE BASE FRAME.

Site	x [m]	y [m]	z [m]
Left foot	-0.03456	0.08940	-0.1935991
Right foot	-0.03456	-0.09180	-0.1935991
Head	0.00600	0.00010	0.2363009

gives

$$\mathbf{p}_L(\mathbf{0}) = \begin{bmatrix} -d_{1x} + d_5 + d_{9x} \\ d_{1y} + d_6 - d_{9y} \\ -d_4 - d_7 - d_8 - d_{9z} \end{bmatrix}, \tag{13}$$

$$\mathbf{p}_R(\mathbf{0}) = \begin{bmatrix} -d_{1x} + d_5 + d_{9x} \\ -d_{1y} - d_6 + d_{9y} \\ -d_4 - d_7 - d_8 - d_{9z} \end{bmatrix}. \tag{14}$$

For the head dimensions in Fig. 4,

$$\mathbf{p}_H(\mathbf{0}) = \mathbf{p}_c + [d_5, -d_3, d_2 + d_4 + d_6]^\mathsf{T}. \tag{15}$$

The resulting site positions are listed in Table IV, and all three zero-pose site rotations are $\mathbf{I}_3$.

III. NUMERICAL INVERSE KINEMATICS

A. Position–orientation objective

For target pose $(\mathbf{p}_t, \mathbf{R}_t)$ and FK prediction $(\mathbf{p}(\mathbf{q}), \mathbf{R}(\mathbf{q}))$, define base-frame residuals

$$\mathbf{e}_p(\mathbf{q}) = \mathbf{p}_t - \mathbf{p}(\mathbf{q}), \tag{16}$$

$$\mathbf{e}_R(\mathbf{q}) = \operatorname{Log}\big(\mathbf{R}_t \mathbf{R}(\mathbf{q})^\mathsf{T}\big)^\vee. \tag{17}$$

Here Log is the principal rotation logarithm, and $\vee$ maps its skew-symmetric matrix to a rotation vector. Its norm is the shortest relative rotation angle, avoiding Euler-angle subtraction. The required dimensionless cost is

$$E(\mathbf{q}) = \|s_p \mathbf{e}_p(\mathbf{q})\|_2^2 + \|\mathbf{e}_R(\mathbf{q})\|_2^2, \qquad s_p = 10 \text{ m}^{-1}. \tag{18}$$

Candidate evaluations use FK only and do not modify MuJoCo state.

B. Finite differences and bounded updates

Let $\mathbf{u}_i$ be the ith coordinate basis vector and set $h = 10^{-6}$ rad. At iteration k, the gradient component is approximated by

$$g_i^{(k)} = \frac{E(\mathbf{q}^{(k)} + h\mathbf{u}_i) - E(\mathbf{q}^{(k)} - h\mathbf{u}_i)}{2h}. \tag{19}$$

TABLE V
MEASURED FK ERRORS OVER 300 RANDOM CONFIGURATIONS. MAXIMA ARE COMPUTED INDEPENDENTLY FOR EACH CHAIN.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	8.450861×10^{-15}	2.285348×10^{-14}	PASS
Right leg	4.429243×10^{-14}	2.357637×10^{-14}	PASS
Head	6.572043×10^{-15}	1.032631×10^{-13}	PASS

TABLE VI
MEASURED IK ERRORS OVER 1,515 TRAJECTORY POSES. MAXIMA ARE COMPUTED INDEPENDENTLY FOR EACH CHAIN.

Chain	Max. position (m)	Max. orientation (rad)	Result
Left leg	9.847942×10^{-6}	6.846493×10^{-5}	PASS
Right leg	9.844913×10^{-6}	7.038121×10^{-5}	PASS
Head	8.393151×10^{-6}	6.773237×10^{-5}	PASS

The fixed-step update is

$$\mathbf{q}^{(k+1)} = \mathrm{clip}\Big(\mathbf{q}^{(k)} - 0.1\mathbf{g}^{(k)}, \mathbf{q}_{\min}, \mathbf{q}_{\max}\Big), \tag{20}$$

with joint limits read from the XML model. The initial guess is the current state of the selected chain. At most 5,000 iterations are allowed per target, with termination when $E < 10^{-8}$.

Since both terms in (18) are nonnegative, the cost criterion implies

$$\|\mathbf{e}_p^{\mathrm{FK}}\|_2 < 10^{-5}\ \mathrm{m}, \qquad \|\mathbf{e}_R^{\mathrm{FK}}\|_2 < 10^{-4}\ \mathrm{rad}. \tag{21}$$

This statement bounds the FK-model residual. It does not guarantee that fixed-step projected gradient descent converges before the iteration limit. Final acceptance is therefore evaluated independently using the MuJoCo site pose after the solved angles have been written to the model.

C. Supplied task-space trajectories

For phase $\phi \in [0, 1]$, the foot displacements are

$$\Delta\mathbf{p}_L(\phi) = 0.020 \sin^2(\pi\phi)(\mathbf{e}_x + \mathbf{e}_z), \tag{22}$$

$$\Delta\mathbf{p}_R(\phi) = 0.020 \cos^2(\pi\phi)(\mathbf{e}_x + \mathbf{e}_z), \tag{23}$$

in metres, with fixed starting orientations. The head follows a reachable neck-centred arc. With $\gamma = 0.1 \sin^2(\pi\phi)$,

$$\mathbf{p}_{t,H}(\phi) = \mathbf{p}_c + \mathbf{R}_y(-\gamma)(\mathbf{p}_{H,0} - \mathbf{p}_c), \tag{24}$$

$$\mathbf{R}_{t,H}(\phi) = \mathbf{R}_y(-\gamma)\mathbf{R}_{H,0}. \tag{25}$$

The bent-leg and head starting configurations are

$$\mathbf{q}_{L,0} = [0, 0, -0.4, 0.8, -0.4]^{\mathsf{T}}, \tag{26}$$

$$\mathbf{q}_{R,0} = [0, 0, 0.4, 0.8, -0.4]^{\mathsf{T}}, \tag{27}$$

$$\mathbf{q}_{H,0} = [0.2, -0.4, 0, 0]^{\mathsf{T}}. \tag{28}$$

The base is fixed at a height of 0.19 m. Subsequent targets are warm-started from the previous solution, including across cycle boundaries; the initial joint angles are not reset each cycle.

IV. EXPERIMENTAL EVALUATION

A. Protocol and error measures

Every comparison uses base-frame site poses. World-frame MuJoCo transforms are converted by

$${}^{B}\mathbf{T}_S^{\mathrm{MJ}} = \big({}^{W}\mathbf{T}_B\big)^{-1}\,{}^{W}\mathbf{T}_S. \tag{29}$$

For two poses $(\mathbf{p}_a, \mathbf{R}_a)$ and $(\mathbf{p}_b, \mathbf{R}_b)$, the reported errors are

$$\epsilon_p = \|\mathbf{p}_a - \mathbf{p}_b\|_2, \tag{30}$$

$$\epsilon_R = \big\|\operatorname{Log}(\mathbf{R}_a\mathbf{R}_b^{\mathsf{T}})^{\vee}\big\|_2. \tag{31}$$

FK is compared at 100 configurations per chain, with each joint sampled uniformly within its limits and random seed zero used for each chain. IK follows five complete cycles, each with 101 phase samples for each of the three chains: 1,515 measured site poses in total.

The stored experiment used Python 3.11.18, NumPy 1.26.4, SciPy 1.14.1, and MuJoCo 3.2.4. The FK, IK, and XML source fingerprints recorded in the experiment manifest still match the current files. The existing measurements are audited and reused for this typeset report; recompiling the report does not rerun the simulation. Headless operation changes only viewer rendering and its delay.

B. Measured results

Tables V and VI report maxima over all stored samples. The required FK limits are $\epsilon_p < 10^{-6}$ m and $\epsilon_R < 10^{-5}$ rad; the IK limits are $\epsilon_p < 10^{-5}$ m and $\epsilon_R < 10^{-4}$ rad. Every recorded value is finite and satisfies its corresponding limit. The largest FK position and orientation errors are 4.429243×10^{-14} m and 1.032631×10^{-13} rad, respectively. The corresponding largest IK errors are 9.847942×10^{-6} m and 7.038121×10^{-5} rad.

Figure 1 shows the full chronological record without dropping initial or boundary samples. Errors near the position threshold are consistent with terminating at the specified weighted-cost criterion; the solver is not required to optimize each target to machine precision. The largest position error remains below the required limit.

C. Regression checks and scope

Six regression test cases additionally cover XML-based zero poses, random configurations under a translated and rotated floating base, the metric weights and relative-rotation angle in the objective, an unchanged target pose, nearby reachable IK targets, and projection onto joint limits. The IK checks also verify that the base and unselected chains are preserved.

The comparison loops update site poses with MuJoCo's `mj_forward` routine, without dynamic stepping. The very small FK errors quantify agreement with the numerical XML model; they do not measure physical link calibration. The random configurations do not exhaust the joint domain. The experiment does not evaluate contact stability, real-robot dynamics, or arbitrary-target IK convergence.

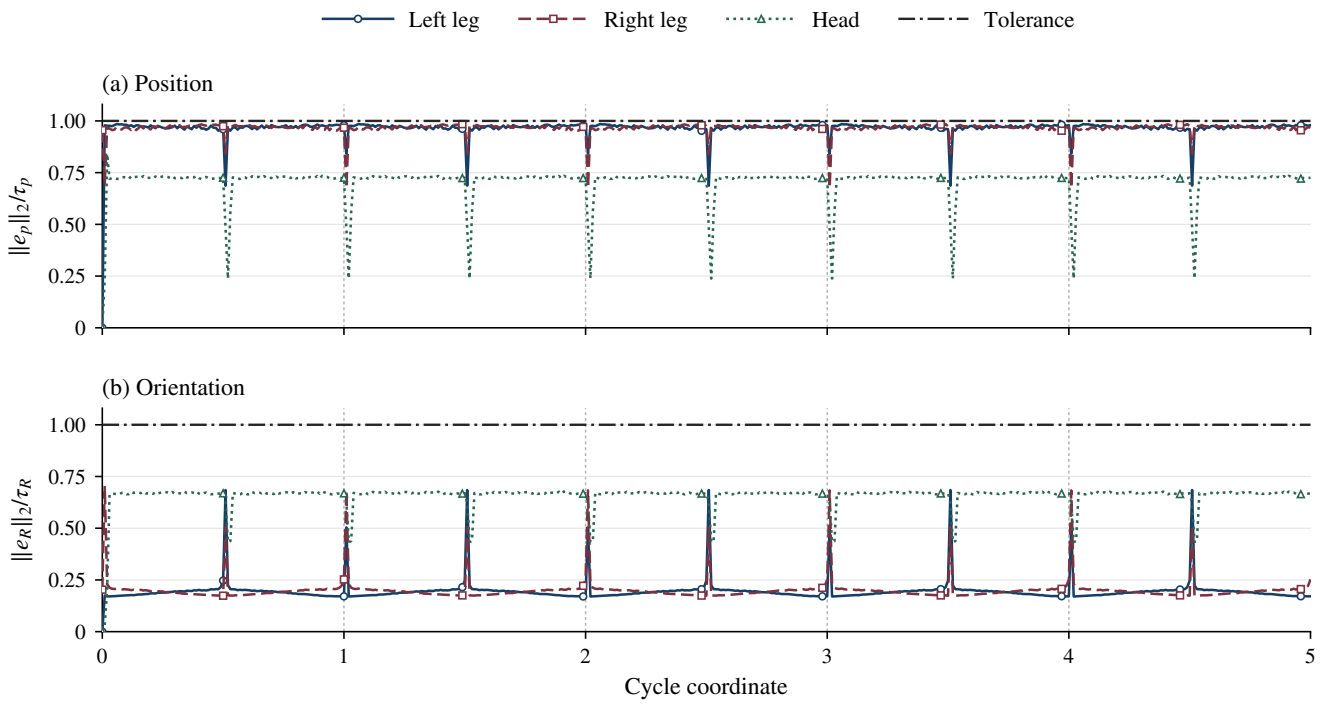


Fig. 1. All measured IK errors, normalized by their acceptance limits: $\tau_p = 10^{-5}$ m for position and $\tau_R = 10^{-4}$ rad for orientation. Each curve contains all 505 targets for one chain, including the initial values; the full comparison contains 1,515 chain poses. An ordinate of one is the acceptance boundary. Vertical lines separate the five cycles. Line styles and markers distinguish the chains in monochrome. No smoothing or resampling is applied; line segments connect adjacent measurements.

V. Conclusion

The derived MDH chains reproduce all three sites of the supplied robot model, including mirrored joint signs and the fixed terminal transforms. The prescribed central-difference, fixed-step IK method satisfies all position and orientation tolerances over the complete supplied trajectory test. The three MDH tables, measured records, source fingerprints, vector figure, and editable LaTeX sources provide a reproducible submission.

Appendix A
Reproduction

The repository includes a pinned Python 3.11 environment, FK and IK entry points, and six regression test cases. Full command lines for the complete headless comparison and JSON exports are included in the supplied README.

The report builder prepares tables from the measured JSON files and compiles the paper with a LaTeX engine. Its manifest records the engine version, compile status, and input fingerprints. The source package contains full command lines, generated tables, a bibliography, and the vector figure, so the document can also be compiled independently.

References

[1] D. T. M. Lau, "Assignment 1: Kinematics," ROSE5710: Fundamental Robotics and Systems, course handout, 2026, first Term, 2026–2027, Figs. 2–4 and Secs. 1–4.

[2] apirrone and contributors, "Open Duck Playground: Open Duck Mini V2 robot model," Source repository. [Online]. Available: https://github.com/apirrone/Open_Duck_Playground, accessed: Nov. 11, 2026. The supplied XML snapshot is identified by SHA-256 in the accompanying experiment manifest.

[3] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 5026–5033.